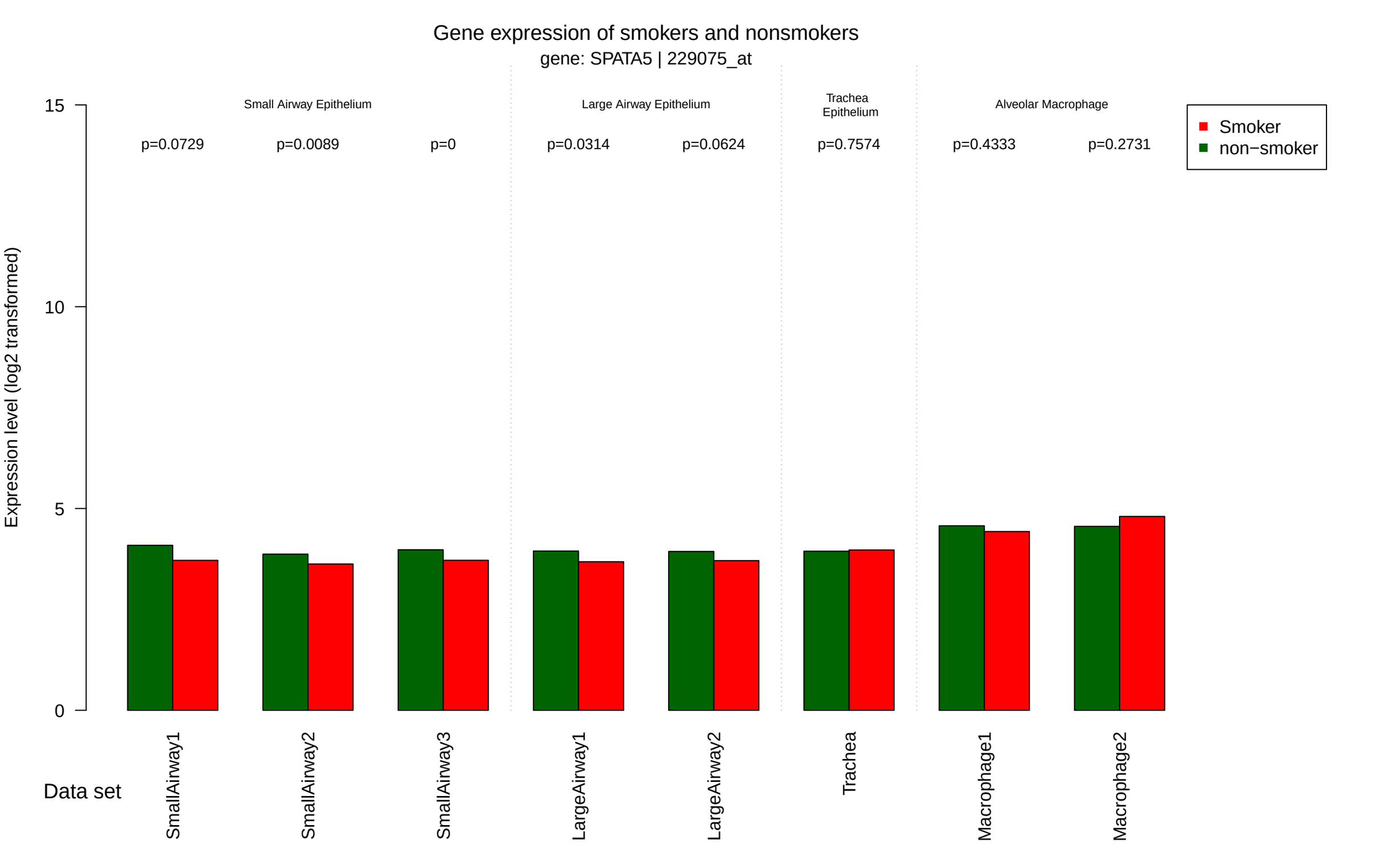
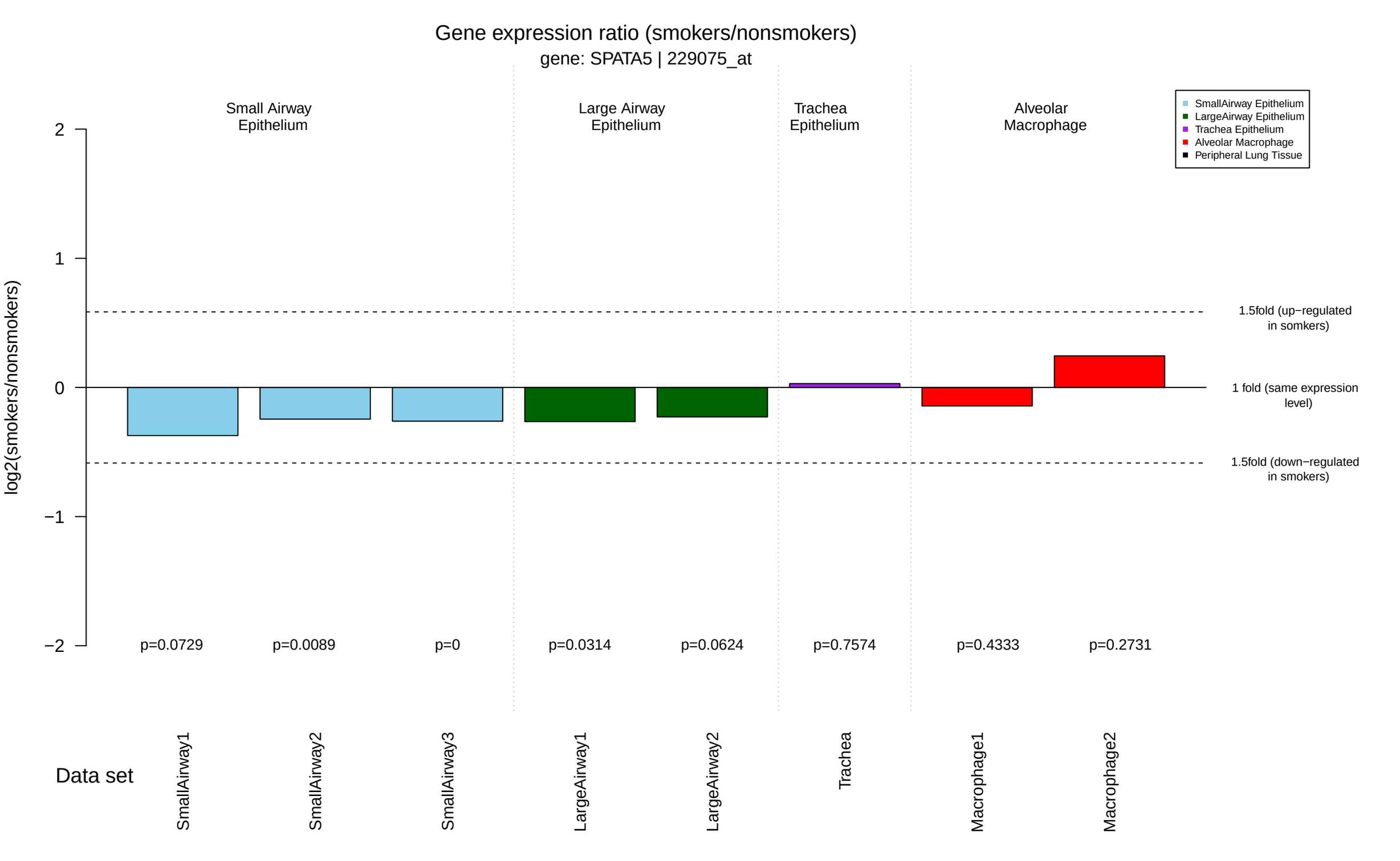
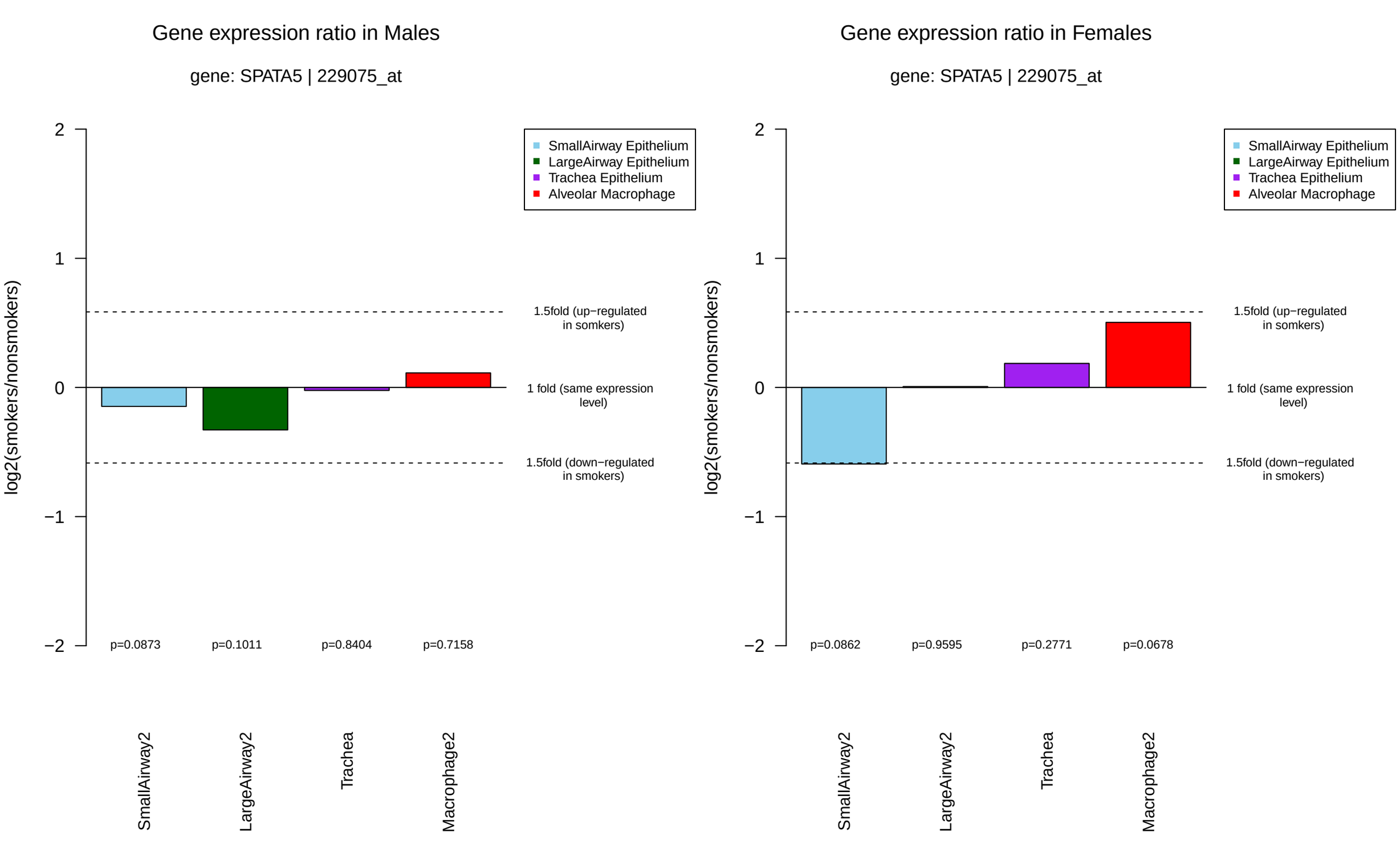
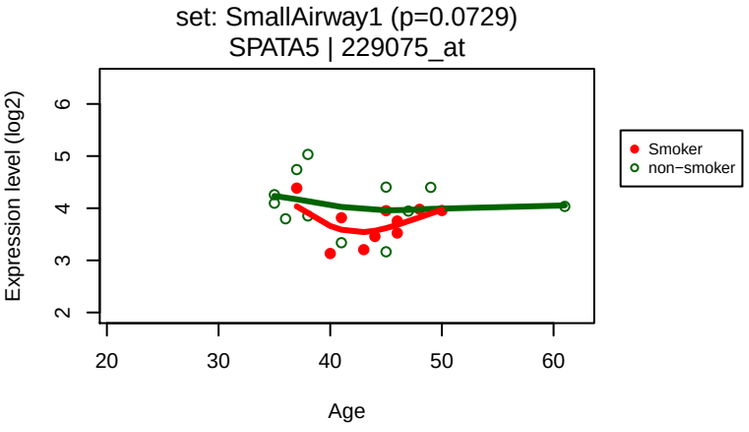
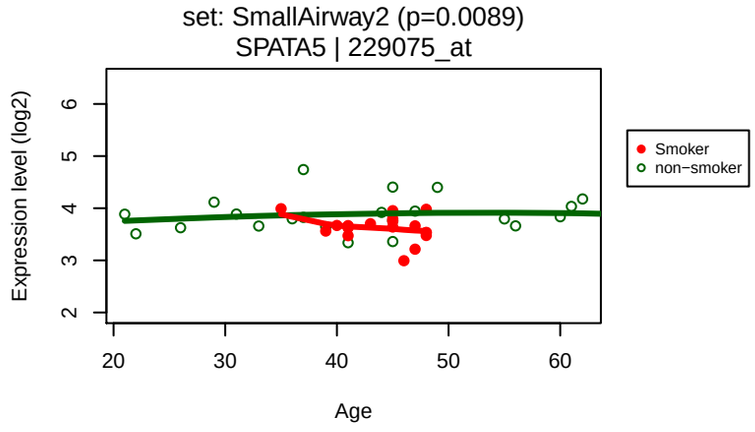
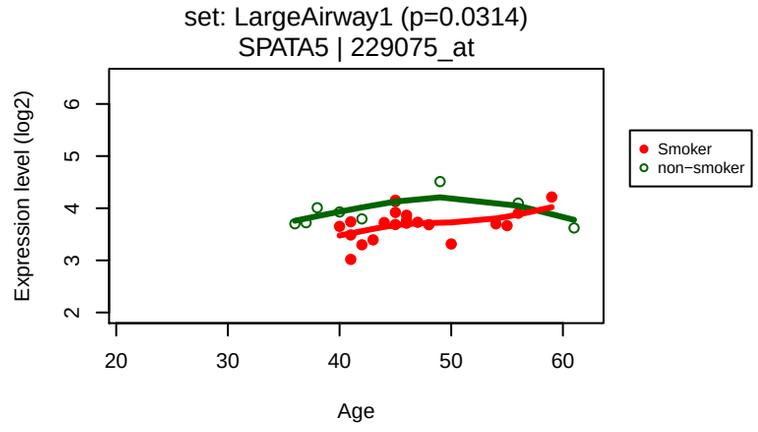

Gene expression of smokers and nonsmokers
gene: SPATA5 | 229075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0729
p=0.0089
p=0
p=0.0314
p=0.0624
p=0.7574
p=0.4333
p=0.2731
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

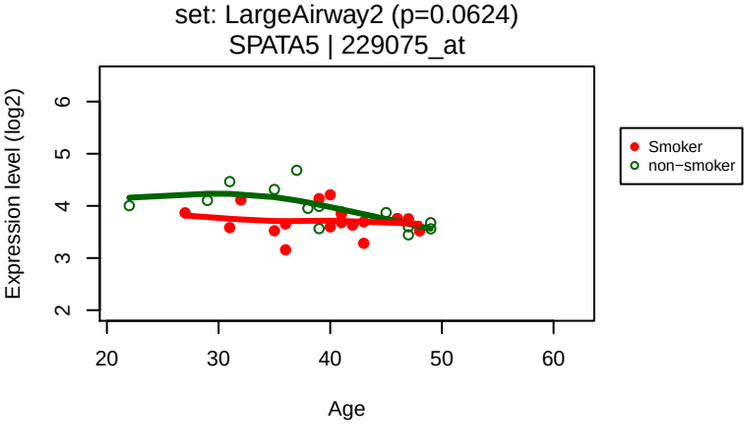

Gene expression ratio (smokers/nonsmokers)
gene: SPATA5 | 229075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0729
p=0.0089
p=0
p=0.0314
p=0.0624
p=0.7574
p=0.4333
p=0.2731
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

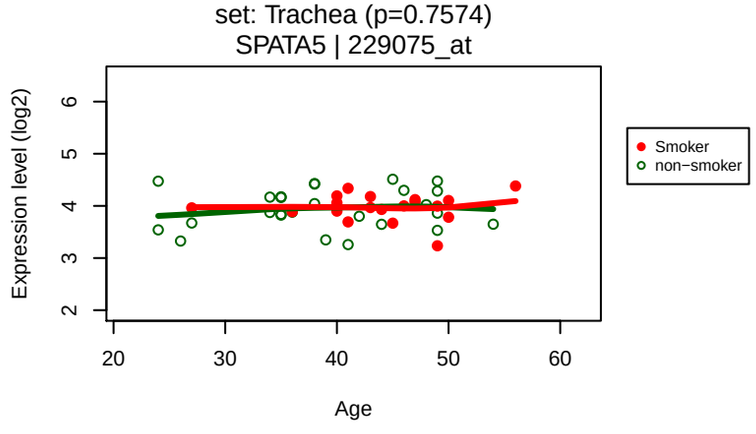

Gene expression ratio in Males
gene: SPATA5 | 229075_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0873
p=0.1011
p=0.8404
p=0.7158
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPATA5 | 229075_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0862
p=0.9595
p=0.2771
p=0.0678
SmallAirway2
LargeAirway2
Trachea
Macrophage2

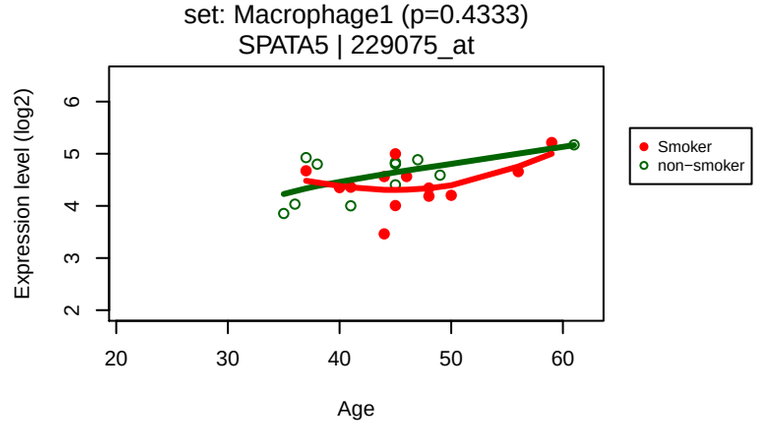

set: SmallAirway1 (p=0.0729)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker

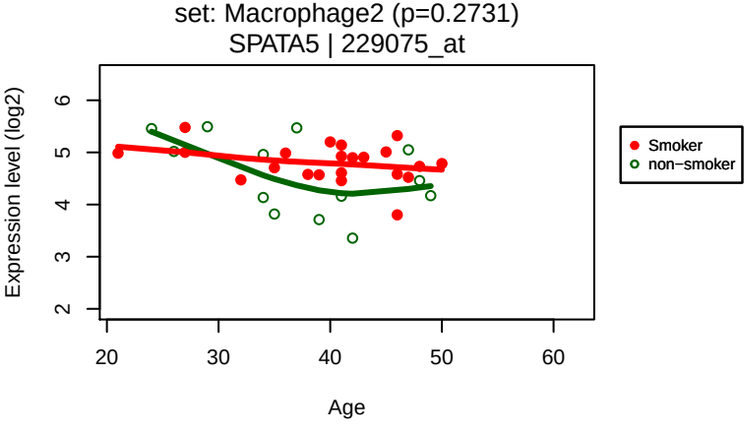

set: SmallAirway2 (p=0.0089)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker

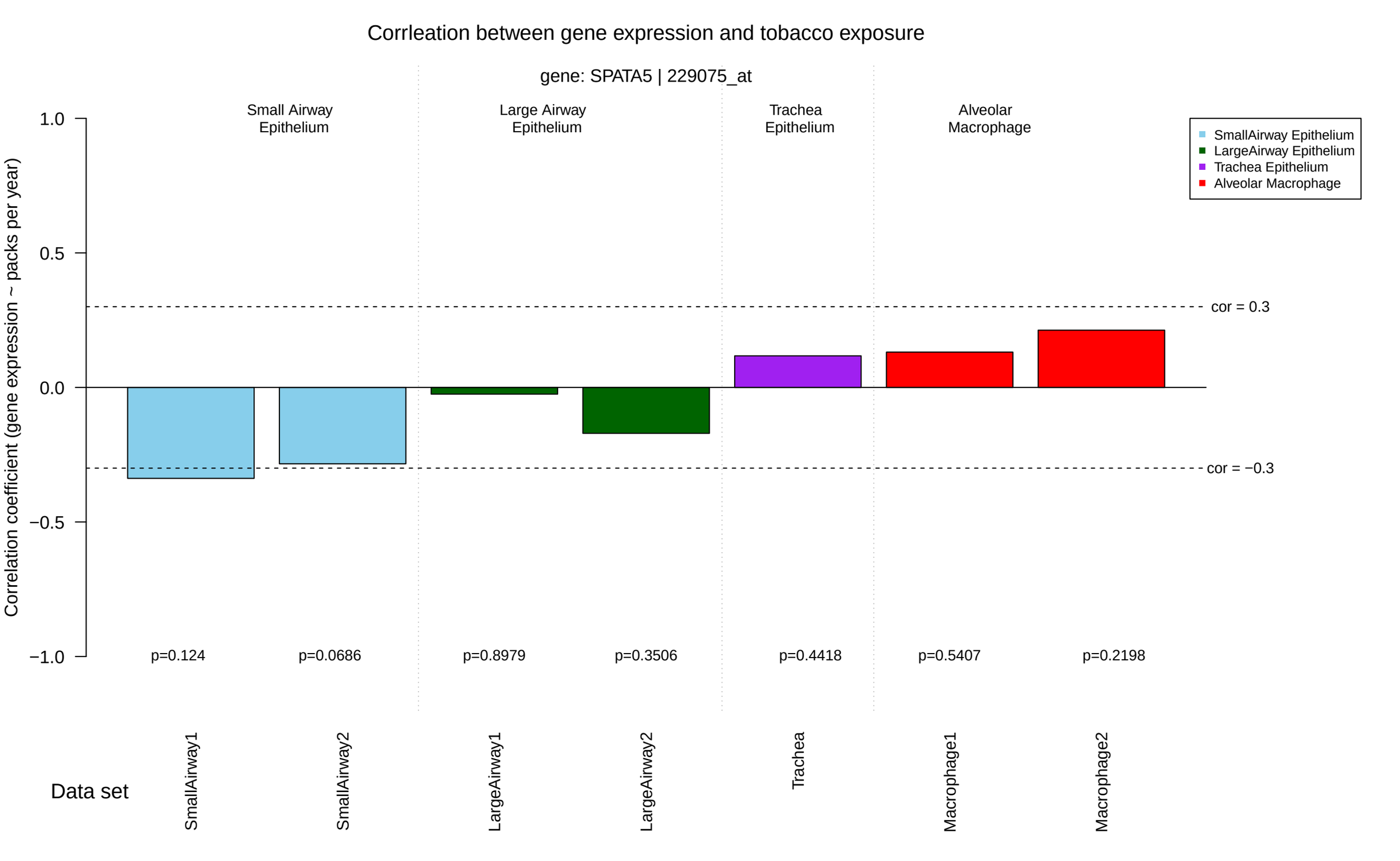

set: LargeAirway1 (p=0.0314)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0624)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7574)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4333)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2731)
SPATA5 | 229075_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPATA5 | 229075_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.124
p=0.0686
p=0.8979
p=0.3506
p=0.4418
p=0.5407
p=0.2198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2